# Band-1 Hadamard order 14596 – Classical Paley-II DIRECT (q2=7297); IN SPK via Paley-II; Gap 2052 Still OPEN

H_14596 Paley-II PASS; Paley-I FAIL/SKIP (q=14595=3*5*7*139 NOT prime); Kronecker FAIL/SKIP (4*3649; 3649=41*89 odd NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

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**Document class.** Scientific computational / construction certificate (Band-1).
**Program.** Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
**Parents.** `2026-10-13_Band1_Hadamard_14592_PaleyI` (current bookkeeping); `2026-10-13_Band1_Hadamard_145` `2026-10-12_Band1_Hadamard_14508_PaleyII` (Paley-II construction + chain template); `2026-10-12_Band1_Hadamard_14520_PaleyI`; `2026-10-09_Band1_Hadamard_14432_PaleyI`; `2026-10-09_Band1_Hadamard_14428_PaleyII`; `2026-10-09_Band1_Hadamard_14356_PaleyII`; `2026-09-21_Band1_Hadamard_9124_PaleyII`; ....
**Lab.** `/workspace/experiments/band1/hadamard_14596_paleyII/`.
**Cutoff.** 2026-10-13T17:12:03Z (~1:12PM ET).

**Honesty stamp (non-negotiable).**
This memo claims **finite construction of order 14596 only (verified max|HHˆT-nI|=0), via classical Paley-II q2=7297 (prime ===1 mod 4), order 2(q2+1). 14596 IN SPK via Paley-II. 14596 is a long-known Hadamard order; this is a finite on-box verification, NOT a new-order claim.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14596: Paley-II **PASS** ($q_2 = 7297$ prime $\equiv 1 \pmod 4$, $n = 2(q_2+1)$); Paley-I **FAIL/SKIP** ($q = 14595 = 3 \cdot 5^2 \cdot 193$ NOT prime); Kronecker **FAIL/SKIP** ($14596 = 4 \cdot 3649$; $3649 = 7 \cdot 11 \cdot 47$ odd composite, NOT Hadamard-admissible; do NOT invent H_3649). **14596** folds into SPK **only** via classical Paley-II (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED** (no invented J). No finished-ToE claim. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14592+14564+14552+14544+14520+14508+14504+14480+14476+14460+14448+14432+14428+144** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

## Abstract

Finite construction of order **14596** (sixth Paley-II DIRECT wave of the 14xxx chain, after 14356, 14428, 14460, 14476 and 14508).

- $H_{14596}$ = Paley-II ($q_2 = 7297$) – classical SPK
- Paley-I for 14596: **FAIL/SKIP** ($q = 14595 = 3 \cdot 5^2 \cdot 193$ NOT prime)
- Kronecker for 14596: **FAIL/SKIP** ($4 \cdot 3649$; $3649 = 7 \cdot 11 \cdot 47$ odd composite, NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

**SPK: IN SPK via Paley-II**.
**Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14596.py`

**Results.** overall_pass=True; units 874/874; row blocks 115/115 (block size 128); wall 7088.0796s; constructed max|HHˆT-nI|=0.

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## Arithmetic

- Target $n = 14596 = 2(q_2 + 1)$ with $q_2 = 7297$ prime $\equiv 1 \pmod 4$ — Paley-II eligible (claim path).
- Paley-I: $q = n - 1 = 14595 = 3 \cdot 5^2 \cdot 193$ NOT prime — FAIL/SKIP.
- Kronecker: $14596 = 4 \cdot 3649$; $3649 = 7 \cdot 11 \cdot 47$ odd composite ($\equiv 3 \pmod 4$) NOT Hadamard-admissible — FAIL/SKIP; no other admissible split.
- Shape: $14596 = 2^2 \cdot 7 \cdot 11 \cdot 47 \equiv 12 \pmod{16}$ (===4 mod 16 / m===1 odd NOT-admissible (shape peer 14356; template 14508 was ===12/m===3) $= 2^2 \cdot 3 \cdot 5 \cdot 241$; prior twin-class $14028 = 2^2 \cdot 3 \cdot 7 \cdot 167$ unbuilt).
- Miyamoto digression skipped this wave (prefer classical SPK).

## Construction (Paley type II)

Let $\chi$ be the quadratic character of $\mathbb{F}_{7297}$ and $Q_{ij} = \chi(j - i)$ (symmetric, since $q_2 \equiv 1 \pmod 4$). The conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & Q \end{pmatrix}$ of order $q_2 + 1 = 7298$ gives

$$H_{14596} = \begin{pmatrix} C + I & C - I \\ C - I & -(C + I) \end{pmatrix}.$$

Built directly in int8 (memory-lean; unit-tested equal to the int64 reference builder on $q \in \{5, 13, 17, 29, 37, 41, 53, 61\}$), then verified block-wise: every row block of $HH^T - nI$ checked exactly in int32.

## Constructions (verified)

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_14596 Paley-II | Paley-II q2=7297 | 0 | 7067.9791 |
| H_14596 Paley-I | FAIL/SKIP (q=14595=3*57*139 NOT prime) | – | – |
| H_14596 Kronecker | FAIL/SKIP (4*3649; 3649=41*89 odd NOT Hadamard-admissible) | – | – |

## SPK closure note

| Quantity | Value |
|---|---|
| 14596 in SPK | **True** (Paley-II q2=7297) |
| Prior Paley-I 14592 | IN SPK via Paley-I q=14591 (~11:06AM ET 2026-10-13) |
| Prior Paley-I 14564 | IN SPK via Paley-I q=14563 (~7:34AM ET 2026-10-13) |
| Prior Paley-II 14508 | IN SPK via Paley-II q2=7253 (~9:01PM ET 2026-10-12) |
| Prior Paley-I 14520 | IN SPK via Paley-I q=14519 (~10:44PM ET 2026-10-12) |
| Prior 14504 | IN SPK via Paley-I (~8:47PM ET 2026-10-09) |
| Prior 14432 | IN SPK via Paley-I (~7:51PM ET 2026-10-09) |
| Prior Paley-II 14428 | IN SPK via Paley-II q2=7213 (~7:10PM ET 2026-10-09) |
| Prior Paley-II 14356 | IN SPK via Paley-II q2=7177 (~1:22PM ET 2026-10-09) |
| Gap 2052 | **OPEN** |
| clay_closed | false |
| GATE-04 | LOCKED |

## Honesty

- clay_closed = false

- GATE-04 LOCKED (no invented J)
- No finished-ToE claim
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- 14596 is a long-known order (Paley 1933 family); finite on-box verification only
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

## Next optional

NEXT direct route after 14596 (FIRST any direct = Paley-II): q2=7309->n=14620 Paley-II DIRECT (prime ===1 mod 4; Paley-I q=14619=3*11*443 composite -> FAIL; Kronecker 14620=4*3655; 3655=5*17*43 ===3 mod 4 NOT Hadamard-admissible -> FAIL/SKIP; template hadamard_14596_paleyII / hadamard_14508_paleyII) | next Paley-I q=14627->n=14628 (template hadamard_14592_paleyI) | Kronecker-only NOTE 14576=4*3644 (H_3644 via Paley-I q=3643; below 14596, note-only, unbuilt) | skipped Paley-II DIRECT orders below 14596 still unbuilt (template hadamard_14596_paleyII / hadamard_14508_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)